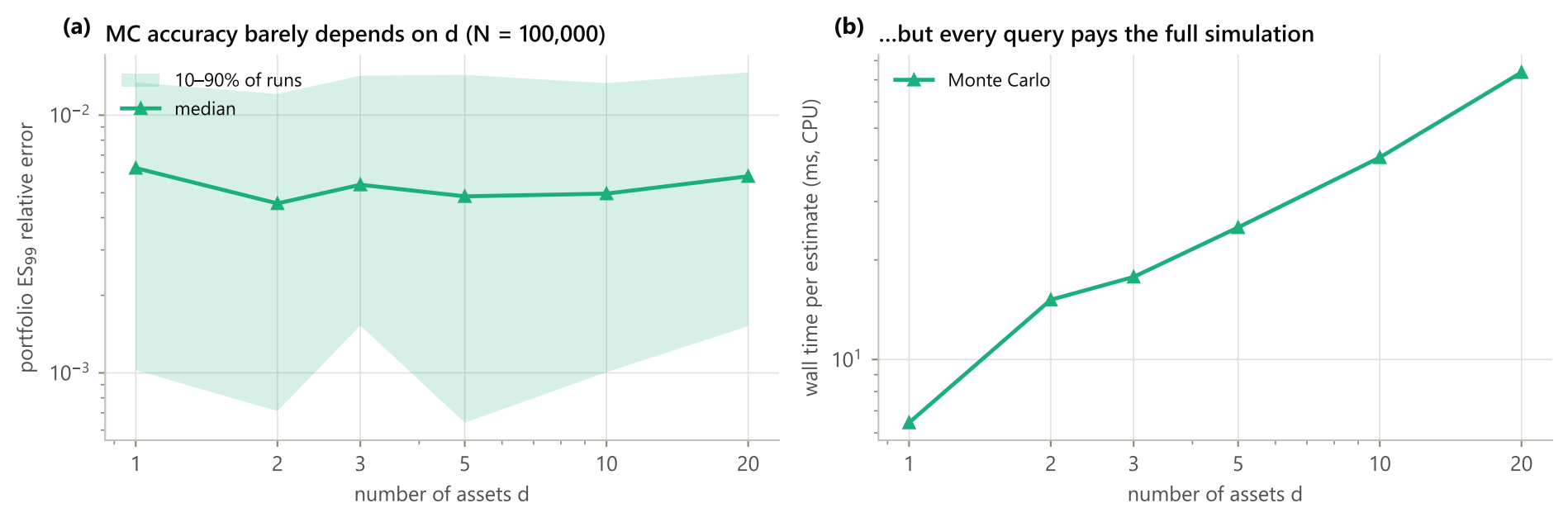
(a)
MC accuracy barely depends on d (N = 100,000)
10–90% of runs
median
10^{-2}
10^{-3}
portfolio ES_{99} relative error
1
2
3
5
10
20
number of assets d
(b)
...but every query pays the full simulation
Monte Carlo
10^{1}
wall time per estimate (ms, CPU)
1
2
3
5
10
20
number of assets d